Exponential and Logarithmic Equations

Exponential equations

1 Solve $3^{x+1} = 81$ exactly.

2 Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.

3 Solve $e^{2x} - 5e^{x} + 6 = 0$ exactly.

Logarithmic equations

4 Solve $\ln(2x - 3) = 1$. Give an exact answer and a decimal to three places.

5 Solve $\log_2 x + \log_2 (x - 6) = 4$, checking for extraneous solutions.

6 Solve $\log(x + 3) - \log(x - 1) = 1$, checking for extraneous solutions.

7 The pH of a solution is $\mathrm{pH} = -\log[\mathrm{H}^+]$. Find the hydrogen-ion concentration $[\mathrm{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.

More exponential equations

8 Solve $2^{x} = 3^{x-1}$, giving an exact expression and a decimal to three places.

9 Solve $4^{x} - 3 \cdot 2^{x} - 4 = 0$ exactly, using the substitution $u = 2^{x}$.

Systems and applications

10 A bacteria culture grows according to $N(t) = 500e^{kt}$. If the population reaches 2000 after 6 hours, find k (4 d.p.) and hence the time for the population to reach 10,000, to one decimal place.

11 Solve the equation $e^{x} + e^{-x} = 3$ exactly, using the substitution $u = e^{x}$ (this expression is related to $2\cosh x$).

Combined logarithmic and exponential practice

12 Solve $\log_3 (x + 2) + \log_3 (x - 2) = 2$, checking for extraneous solutions.

13 Solve $9^x = 27^{x-1}$ exactly, writing both sides with base 3.

Equations requiring both exponential and logarithmic techniques

14 Solve $\log_2(x) = 3 - \log_2(x-2)$, checking for extraneous solutions.

15 Solve $e^{\ln x + \ln 2} = 10$ for x, simplifying the exponent first.

Verifying solutions numerically

16 For the equation $5^{2x} = 17$ solved earlier ($x \approx 0.880$), verify the solution by substituting back and computing both sides to 2 decimal places.

Equations with different bases on each side

17 Solve $7^x = 3^{x+2}$, giving an exact expression and a decimal to three places.

18 Solve $10^{2x-1} = 500$, giving an exact logarithmic expression and a decimal to three places.

A modeling problem requiring both equation types

19 A population model is $P(t) = 800e^{0.05t}$. Find the time t when $P(t) = 2000$, to two decimal places, and separately find $P^{-1}(1500)$ (the time when the population reaches 1500), to two decimal places.

Checking solutions by substitution

20 Verify the solution $x = 8$ found earlier for $\log_2 x + \log_2(x-6) = 4$ by substituting back into the original equation.

21 Solve $\ln(x) + \ln(x-3) = \ln(10)$, checking for extraneous solutions.

22 A radioactive sample decays according to $A(t) = A_0e^{-0.08t}$. If the initial amount is 500 g, find the time t (to two decimal places) at which only 50 g remains, and separately find the time at which 90% of the original sample has decayed away.

23 Solve $4^x = 8^{x-1}$ exactly, writing both sides with base 2.

24 Solve $\log_5(x+4) = 2$ for x.